

Superior piezoelectric performance in holmium-substituted high- T_C $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$

Hui-Lin Li^a, Yi-Nuo Chen^b, Qian Wang^a, Chun-Ming Wang^{a,c,*}

^a School of Physics, State Key Laboratory of Crystal Materials, Shandong University, Jinan, Shandong 250100, China

^b Jinan University Town Experimental Senior High School, Jinan, Shandong 250306, China

^c Center for Optics Research and Engineering (CORE), Key Laboratory of Laser and Infrared System of Ministry of Education, Shandong University, Qingdao 266237, China

ARTICLE INFO

Keywords:

Bismuth layer-structured ferroelectrics (BLSFs)

Bismuth titanate-ferrite ($\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$)

Holmium oxide (Ho_2O_3)

Curie temperature (T_C)

High-temperature piezoelectric ceramics

ABSTRACT

High-temperature piezoelectric materials are essential for advanced sensor technologies, yet achieving both superior piezoelectric properties and a wide operating temperature range remains a persistent challenge. Bismuth titanate-ferrite ($\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$) is a promising candidate due to its notably high Curie temperature (T_C) of approximately 761 °C. However, its low intrinsic piezoelectric response and limited poling efficiency, attributed to insufficient direct-current (dc) resistivity, have restricted its broader utilization. To address these issues, we explored compositional modifications by partially substituting A-site bismuth ions with holmium ions ($\text{Bi}_{5-x}\text{Ho}_x\text{Ti}_3\text{FeO}_{15}$, BTF-100xHo), aiming to enhance both piezoelectric performance and electrical resistivity. Comprehensive characterization was performed on the holmium-substituted ceramics, examining their crystal structure, microstructural attributes, and piezoelectric properties. Our findings reveal that the holmium-substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ compositions exhibit significantly improved piezoelectric behavior in comparison to their unmodified counterparts. The optimized BTF-4Ho composition demonstrates a high piezoelectric coefficient ($d_{33} = 23.1$ pC/N) and an elevated T_C of 795 °C. Moreover, the BTF-4Ho composition maintains stable electromechanical coupling across a wide temperature range, underscoring its potential for high-temperature sensor applications.

1. Introduction

Over the past many years, piezoelectric ceramics have gained increasing interest and spurred extensive research due to their significant potential in developing new functional devices [1–3]. The dominant commercial piezoelectric ceramics are lead-based, represented by lead zirconate-titanate (PZT) [4–6]. However, the relatively low Curie temperature (T_C) of ~360 °C for PZT limits its applicability in high-temperature environments such as aerospace, automotive electronics, and geological exploration [7–9]. Therefore, there is an urgent need to investigate high-performance high- T_C piezoelectric ceramics that can be operated at high temperatures.

In a variety of high-temperature piezoelectric ceramics, bismuth layer-structured ferroelectrics (BLSFs) have been widely studied for their high T_C , large spontaneous polarization (P_s), and low dielectric loss [10–14]. Bismuth titanate-ferrite ($\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$) is a typical four-layer ($m = 4$) BLSF compound. Structurally, $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ consists of a four-layer

intergrown ABO_3 perovskite unit of $(\text{Bi}_3\text{Ti}_3\text{FeO}_{13})^{2-}$ sandwiched by a $(\text{Bi}_2\text{O}_2)^{2+}$ unit along the c -axis [15–18]. In this structure, Bi^{3+} ions in $(\text{Bi}_3\text{Ti}_3\text{FeO}_{13})^{2-}$ unit occupy the A-sites, while Ti^{4+} and Fe^{3+} ions are located at the B-sites with probabilities of 75 % and 25 %, respectively [19]. The multiferroic, photocatalytic, and ferroelectric photovoltaic properties of BTF have been extensively investigated [19–22]. However, its piezoelectric properties, especially at high temperatures, have not been sufficiently explored. Moure et al. reported an enhanced piezoelectric constant (d_{33}) of 18.6 pC/N in $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ - $\text{CaBi}_4\text{Ti}_4\text{O}_{15}$ (BTF-CBT), but found that the d_{33} of pure $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ is 0 pC/N [23]. Similarly, Huang et al. achieved a d_{33} of 13.4 pC/N in (Na, Ce) co-modified $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$, but no piezoelectricity was found in pure $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ [24]. Nevertheless, the inherent ferroelectricity of $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ indicates that it should exhibit piezoelectricity [25]. Consequently, we conducted research on $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$, and we obtained a d_{33} of 7 pC/N for pure $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ and an enhanced d_{33} of 23 pC/N for manganese (Mn)-substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$, with the T_C of BTF-4Mn (765

* Correspondence to: School of Physics, Shandong University, Jinan, Shandong 250100, China.

E-mail address: wangcm@sdu.edu.cn (C.-M. Wang).

<https://doi.org/10.1016/j.mtcomm.2024.111118>

Received 20 October 2024; Received in revised form 20 November 2024; Accepted 24 November 2024

Available online 26 November 2024

2352-4928/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

°C) remaining almost unchanged compared to pure $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ (761 °C) [25]. We also achieved a high d_{33} of 24.3 pC/N for tungsten (W)-substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$, with a T_C of 793 °C [17]. According to previous works, substituting rare-earth ions with smaller ionic radii at the A-site of BLSFs significantly influences T_C and piezoelectric properties. For example, substituting Tb^{3+} for Ca^{2+} in $\text{CaBi}_2\text{Nb}_2\text{O}_9$ increased T_C from 939 °C to 973 °C as Tb^{3+} content increased from 0 to 0.07 mol [26]. Holmium (Ho), a rare-earth element, plays a pivotal role in enhancing the electrical properties of ferroelectric oxides. Recent research has demonstrated that doping ferroelectric ceramics such as $\text{SrBi}_4\text{Ti}_4\text{O}_{15}$, $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$, and perovskite-type $\text{Bi}_{0.5}\text{Na}_{0.5}\text{TiO}_3$ with Ho ions can significantly improve their piezoelectric performance [27–29]. Therefore, we fabricated holmium (Ho)-substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics using the conventional solid-state reaction method. We investigated the effects of Ho^{3+} substitution on the structural, dielectric, piezoelectric, and electrical properties of $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics in detail.

2. Experimental procedure

$\text{Bi}_{5-x}\text{Ho}_x\text{Ti}_3\text{FeO}_{15}$ (BTF-100xHo, $x = 0.00, 0.02, 0.03, 0.04, 0.05, 0.06, 0.08, 0.10$) ceramics were synthesized using the conventional solid-state reaction method. Analytical grade Bi_2O_3 (99.8 %), Fe_2O_3 (99.4 %), TiO_2 (99.8 %), and Ho_2O_3 (99.99 %) were utilized as raw materials. The weighed chemicals were wet-milled in polyethylene bottles with ZrO_2 balls for 12 h in ethanol, followed by calcination at 750 °C for 3 hours. The calcined powders were then milled, dried, ground, and granulated with a polyvinyl alcohol (PVA) binder. The granulated powders were pressed into discs with diameters of 12.5 mm and thicknesses of 1.0 mm at a pressure of 150 MPa. The pellets were subsequently heated to 650 °C for 60 min to burn out the binder. Finally, the samples were sintered at 1080 °C for 3 h. The sintered samples were polished to a thickness of approximately 0.5 mm, and silver paste was screen-printed onto the two parallel polished surfaces, followed by firing at 570 °C for 30 min.

The phase structure was characterized using an X-ray diffractometer (Smartlab SE, Rigaku Corporation, Tokyo) with $\text{Cu K}\alpha$ radiation in the 2θ range of 20°–70°. Microstructural characterization of the sintered samples was performed using field-emission scanning electron microscopy (FE-SEM, Carl Zeiss, Merlin Compact). For electrical and dielectric property measurements at room temperature, silver electrodes were affixed to both surfaces of the sintered pellets and fired at 570 °C for 20 minutes in air. The samples were poled in silicone oil for 20 minutes under a dc electric field at 150–200 °C. The piezoelectric constant d_{33} was measured using a quasistatic d_{33} meter (Institute of Acoustics, Academia Sinica, ZJ-2). Dielectric spectra measurements were conducted with a 4294 A impedance analyzer. The planar electromechanical coupling factor k_p was calculated according to Onoe's equation [30]. Ferroelectric hysteresis loops were measured using a TF Analyzer 2000E Measurement System (aixACCT Systems GmbH).

3. Results and discussion

Fig. 1 presents the X-ray diffraction (XRD) patterns of the BTF-100xHo ceramics. All diffraction peaks are accurately indexed to the standard PDF card (PDF#89–8545), confirming the presence of a single orthorhombic phase with the space group $A2_1am$. No impurity phases were detected in any of the samples, suggesting successful incorporation of holmium dopant ions into the pseudo-perovskite layers, occupying the A-site (i.e., Bi^{3+} site) due to their similar effective ionic radii. The strongest diffraction peak, indexed as (119), aligns with the $(112m+1)$ reflection, which corresponds to the strongest diffraction peak of BLSFs [31–34]. The diffraction peaks are located in nearly identical positions across different holmium contents, indicating no significant changes in the XRD patterns of holmium-substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$. This suggests that the crystal lattice parameters remain almost unchanged with increasing holmium content, which can be attributed to the similar ionic radii of

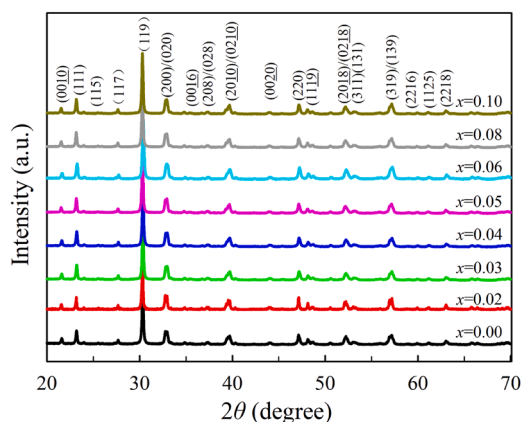


Fig. 1. X-ray powder diffraction (XRD) patterns of the BTF-100xHo ceramics.

Ho^{3+} (124 pm, 12CN) and Bi^{3+} (131 pm, 12CN). Therefore, the introduction of holmium ions primarily affects the local bond structure without altering the overall lattice structure.

Fig. 2 presents the scanning electron microscopy (SEM) images of the as-synthesized $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$, BTF-4Ho, and BTF-8Ho. These images reveal a low-porosity microstructure, confirming the high density of the samples. The SEM images prominently display plate-like grains, a typical feature of BLSF ceramics. All micrographs exhibit a distinct plate-like morphology, with highly anisotropic grains; the in-plane grain size is significantly larger than the thickness along the c -axis. This anisotropy is attributed to the high grain growth rate perpendicular to the c -axis, a unique characteristic of BLSFs [35]. Additionally, the SEM images show that the holmium substitution in $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ leads to larger grain sizes, which reduces the formation of grain boundaries and defects, thereby enhancing the piezoelectric properties [25]. The dense, pore-free microstructure further indicates that these ceramics can withstand large electric fields, which is beneficial for their dielectric, ferroelectric, and piezoelectric performance.

Fig. 3 presents the temperature-dependent dielectric constant ϵ_r and dielectric loss $\tan\delta$ of the BTF-100xHo, measured at a frequency of 1 MHz. A single significant dielectric anomaly is detected in the measured temperature range, representing the ferroelectric-paraelectric phase transition at the T_C . The inset of Fig. 3(A) shows the temperature dependence of the dielectric constant near T_C . The T_C of $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ is 761 °C, which is consistent with previously reported data [24]. The physical parameters of the dielectric properties of BTF-100xHo are listed in Table 1. T_C increases from 761 °C for pure $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ to 796 °C for BTF-8Ho. This change in T_C is closely related to the tolerance factor (t), as numerous studies have shown that a decrease in t often corresponds to an increase in T_C [36]. The tolerance factor is defined as $t = (R_A + R_O) / \sqrt{2(R_B + R_O)}$, where R_A , R_B , and R_O are the ionic radii of the A-site cation, B-site cation, and oxygen ion, respectively. Replacing Bi^{3+} with Ho^{3+} , which has a smaller ionic radius, reduces the parameter t and results in an increase in T_C . Similarly, doping the A-site in other bismuth-based layered materials follows this pattern [37]. Fig. 3(B) also shows the temperature-dependent dielectric loss $\tan\delta$ of the BTF-100xHo. The introduction of holmium ions effectively reduces the dielectric loss at high temperatures. In the wide temperature range of room temperature to 500 °C, the dielectric loss $\tan\delta$ remains quite low. However, once the temperature exceeds 600 °C, the increase in high-temperature conductivity and the thermal activation of domain wall motion dramatically increase the dielectric loss $\tan\delta$ [32]. The inset of Fig. 3(B) presents the dielectric loss $\tan\delta$ on a logarithmic scale to visually demonstrate the variation trend. The composition of BTF-8Ho exhibits the lowest dielectric loss $\tan\delta$ among the holmium-substituted BTF ceramics.

For effective comparison, Fig. 4 shows the T_C and d_{33} of reported

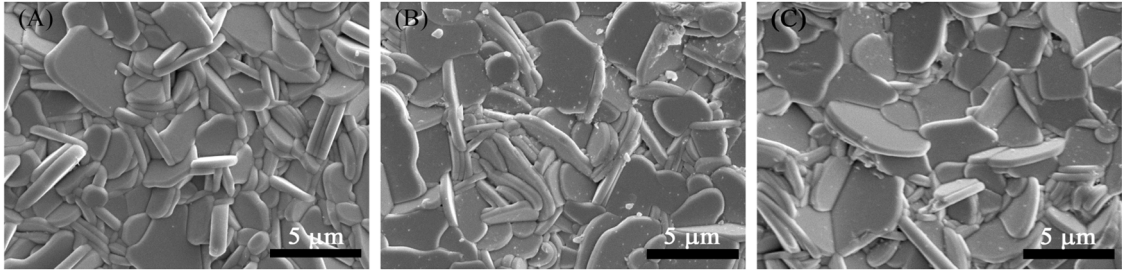


Fig. 2. Scanning electron microscopy images showing the surfaces of (A) $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$, (B) BTF-4Ho, and (C) BTF-8Ho ceramics.

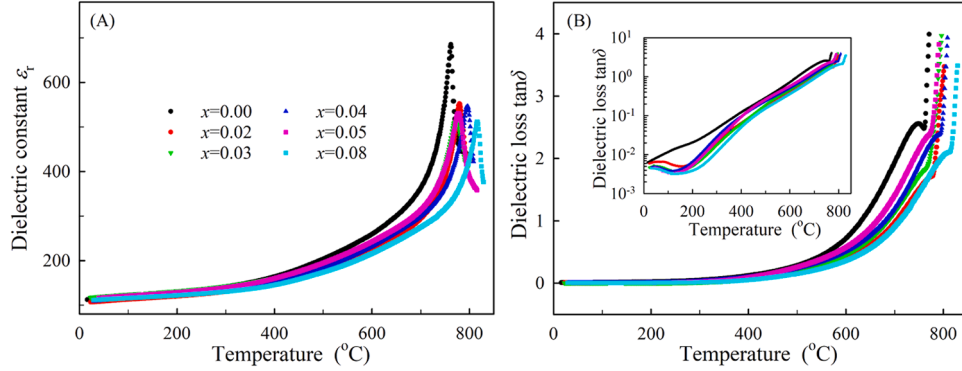


Fig. 3. Temperature-dependent (A) dielectric constant ϵ_r , and (B) dielectric loss $\tan\delta$ of the BTF-100xHo, measured at 1 MHz.

Table 1
Room temperature dielectric and piezoelectric properties of BTF-100xHo ceramics.

Compositions	ϵ_r^a	$\tan\delta^a$ (%)	d_{33} (pC/N)	T_C (°C)	N_p (Hz·m)	k_p (%)	Q_{mp}
$\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$	114	0.58	7.1	760	2053	4.6	1877
BTF-2Ho	106	0.59	21.8	779	1996	4.3	2080
BTF-3Ho	117	0.48	21.1	782	2127	3.8	2948
BTF-4Ho	108	0.46	23.1	795	2014	5.2	2026
BTF-5Ho	116	0.45	20.0	795	2112	4.1	2359
BTF-8Ho	111	0.44	18.6	796	1964	4.5	1844

^a Data measured at 1 MHz.

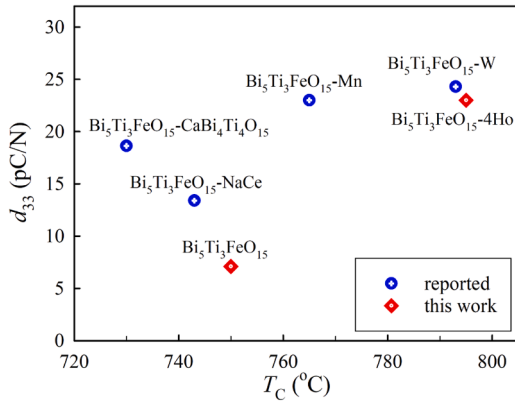


Fig. 4. Comparison of T_C and d_{33} of the reported $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ based ceramics [17,23–25].

$\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ based ceramics [17,23–25]. In the context of comparison with other $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ based materials, this study has achieved a significant enhancement in the T_C , reaching up to 795 °C for BTF-4Ho, as well as a substantial increase in the piezoelectric constant d_{33} , reaching

23.1 pC/N. These results highlight the potential of BTF-4Ho for high-temperature piezoelectric applications.

Fig. 5 presents the polarization-electric field (P - E) hysteresis loops and electric current-electric field (J - E) loops of $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$, BTF-4Ho, and BTF-8Ho ceramics measured at 150 °C and at a frequency of 1 Hz at a drive electric field of 190 kV/cm. All $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics exhibit typical ferroelectric P - E loops, and the switching electric current peak provides a more accurate evaluation of the ferroelectric properties of the materials [38–40], see Fig. 5(B). The current peaks shift to higher drive electric fields with increasing holmium content, indicating that the coercive field (E_c) increases with higher holmium content. Concurrently, the remnant polarization (P_r) decreases with an increase in holmium content. The P_r and E_c of $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ are 11.8 $\mu\text{C}/\text{cm}^2$ and 63.8 kV/cm, respectively. In comparison, the P_r and E_c of BTF-4Ho are 6.6 $\mu\text{C}/\text{cm}^2$ and 65.4 kV/cm, and for BTF-8Ho, they are 9.4 $\mu\text{C}/\text{cm}^2$ and 71.4 kV/cm. The ferroelectric properties of $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ decrease after holmium (Ho) doping, which we attribute to an increase in oxygen vacancies. These vacancies hinder the switching of local polarization near their sites, leading to an increase in both the coercive field and the saturation electric field, as well as a decrease in residual polarization. Consequently, the overall ferroelectric performance is reduced [41]. Although the decrease in P_r (or P_s) seems inconsistent with the increase in the piezoelectric performance, herein d_{33} , the polarization of ferroelectrics originates from the alignment of ferroelectric domains parallel to the poling field [42–44]. Therefore, there is no inevitable correlation between ferroelectric properties and piezoelectric properties [45].

Fig. 6(A) shows the temperature dependence of dc electrical resistivity for the BTF-100xHo in the temperature range of 200–450 °C. Maintaining a high electrical resistivity during poling is crucial as it allows for the application of a substantial electric field without the risk of electrical breakdown or significant charge leakage. The resistivity of all samples decreases with increasing temperature, indicating a negative temperature coefficient of resistance. The pure and holmium-substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics show high resistivity of $>10^5 \Omega\cdot\text{cm}$ up to 450 °C. Moreover, the holmium-substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramic exhibits a lower resistivity compared to the pure $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramic. The

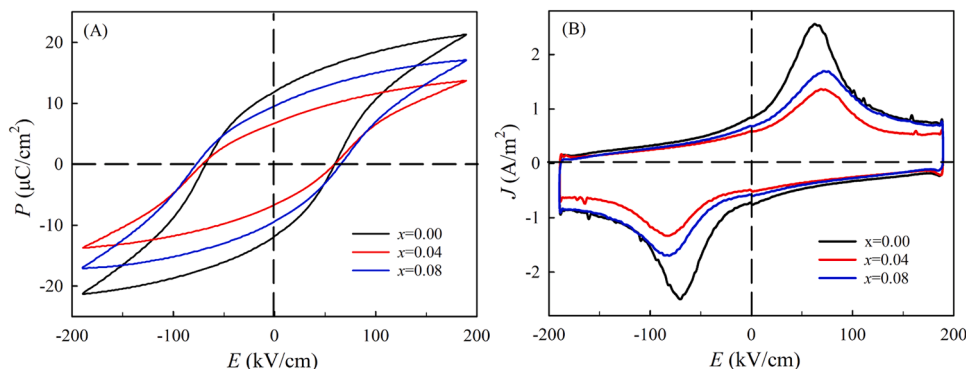


Fig. 5. (A) polarization-electric field (P - E) hysteresis loops and (B) electric current-electric field (J - E) loops of the BTF-100xHo ceramics.

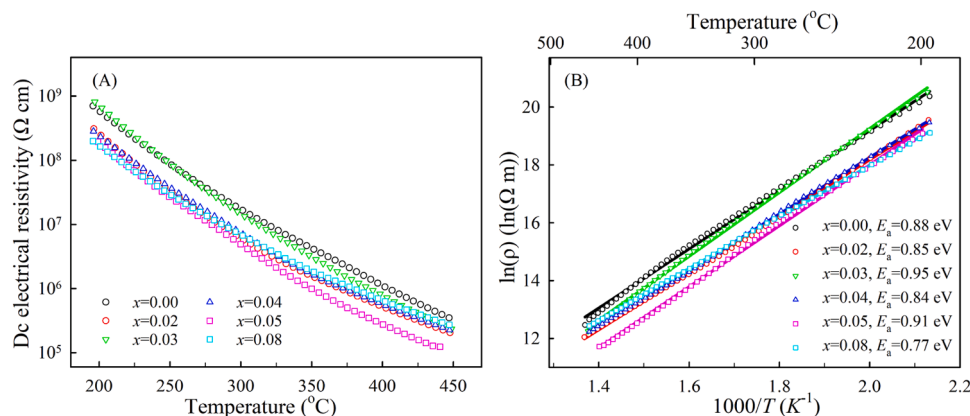


Fig. 6. (A) Temperature dependence of dc electrical resistivity, and (B) the Arrhenius relationship between dc electrical resistivity and temperature for the BTF-100xHo ceramics.

decrease in the resistivity related to the defects is probably one of the reasons of the enhancement of piezoelectric activities. Similar phenomena have been reported in other ions substituted BLSFs [46,47]. To further investigate its conductivity, Fig. 6(B) shows the activation energy (E_a) according to the Arrhenius law, given by $\rho = \rho_0 \exp(-E_a/k_B T)$. Here, ρ_0 is a pre-exponential factor, E_a is the activation energy, k_B is the Boltzmann constant, and T is the thermodynamic temperature. The E_a of intrinsic conduction for $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ based ceramics should be <1.6 eV [48–50]. The activation energy E_a for the samples, which falls between 0.77 and 0.95 eV, is comparable to the values for thermal-activated oxygen vacancies [51].

Table 1 summarizes the dielectric, piezoelectric, and electromechanical properties of BTF-100xHo ceramics at room temperature. Compared with pure $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$, all Ho-substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics exhibit larger d_{33} values. The d_{33} values increase with increasing holmium content up to $x=0.04$ and then decrease with further increases. BTF-4Ho exhibits the maximum d_{33} value of 23.1 pC/N, which is three times higher than that of pure $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$. These results indicate that the piezoelectric performance of $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ can be effectively enhanced by substituting Ho^{3+} for Bi^{3+} at the A-site. The T_C of BTF-4Ho has been measured to be 795 °C, representing an improvement over the pure $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$, and compared with the pure $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$, the introduction of holmium ions reduces the dielectric loss $\tan\delta$. However, the values of the planar frequency constant (N_p) remain almost unchanged with holmium substitution. The k_p of substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ increases from 4.6 % to 5.2 % for BTF-4Ho. The planar mechanical quality factor (Q_{mp}) significantly increases from 1877 to 2948 with holmium substitution.

To study the thermal stability of the piezoelectric properties of holmium-substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics, the planar

electromechanical coupling coefficient (k_p) was measured at various temperatures. Fig. 7 shows the frequency-dependent electrical impedance ($|Z|$) of BTF-4Ho in planar mode at different temperatures. The resonance amplitude gradually decreases as the temperature rises from 20 °C to 380 °C, and then rapidly decreases with further temperature increase, which can be attributed to the increase in conductance at high temperatures [52]. The experimental results demonstrate that the hardness of the materials diminishes as the temperature rises [53]. At 380 °C, the amplitude of the resonant peak remains relatively high, indicating a good resonance and anti-resonance response. This suggests that the resonance characteristics of BTF-4Ho exhibit excellent thermal stability below 380 °C. Additionally, as the temperature increases, the resonant frequency shifts to lower frequencies. Based on the impedance spectrum's temperature variations, the resonant frequency (f_r) and anti-resonant frequency (f_a) of BTF-4Ho in k_p mode are depicted in the inset of Fig. 7(b). Below 400 °C, the resonant frequency shows an almost linear decline trend, indicating promising potential for resonator applications [54].

4. Conclusions

$\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics with holmium substitution are synthesized using conventional solid-state processing, and the effects of holmium on the structural, dielectric, piezoelectric, ferroelectric, and electrical properties of $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics were investigated in detail. X-ray diffraction (XRD) results revealed the crystal structure of the $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics, confirming the successful incorporation of holmium into the $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ lattice. Scanning electron microscopy (SEM) micrographs indicated a significant increase in grain size for the holmium-substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics. The piezoelectric and dielectric

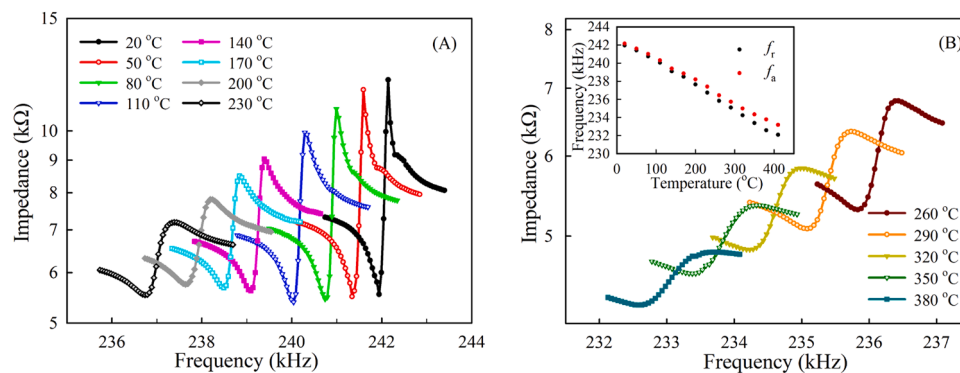


Fig. 7. (A) and (B) Frequency-dependent electrical impedance $|Z|$ of BTF-4Ho in the planar mode at different temperature.

properties of the $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics were notably enhanced by holmium substitution, with BTF-4Ho demonstrating superior performance, featuring a piezoelectric constant d_{33} of 23.1 pC/N and a high Curie temperature T_C of 795 °C. Additionally, BTF-4Ho exhibited low dielectric losses at elevated temperatures and displayed remarkable thermal stability in its piezoelectric response and electromechanical coupling characteristics. These outstanding features suggest that holmium-substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics are promising candidates for high-temperature piezoelectric applications.

Ethical approval

This study does not contain any human-related information and no animals were used. This work did not require any ethical approval. The work is completely original based on the research carried out by authors.

CRediT authorship contribution statement

Hui-Lin Li: Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Yi-Nuo Chen:** Investigation, Data curation. **Qian Wang:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition. **Chun-Ming Wang:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant Number: 52372115), and the Fundamental Research Funds for the Central Universities. Dr. Q.W. also acknowledges financial support from the State Key Laboratory of Crystal Materials, Shandong University (Grant No. KF2308). The authors acknowledge the assistance of the Core Facilities Sharing Platform of Shandong University. The authors also thank the National Demonstration Center for Experimental Physics Education (Shandong University) for X-ray diffraction measurements.

Consent for publication

All authors declare their consent for publication of the manuscript.

Data Availability

Data will be made available on request.

References

- [1] T. Takenaka, H. Nagata, Current status and prospects of lead-free piezoelectric ceramics, *J. Eur. Ceram. Soc.* 25 (2005) 2693–2700.
- [2] T.-L. Zhao, A.A. Bokov, J. Wu, H. Wang, C.-M. Wang, Y. Yu, C.-L. Wang, K. Zeng, Z.-G. Ye, S. Dong, Giant piezoelectricity of ternary perovskite ceramics at high temperatures, *Adv. Funct. Mater.* 29 (2019) 1807920.
- [3] D. Damjanovic, Materials for high temperature piezoelectric transducers, *Curr. Opin. Solid State Mater. Sci.* 3 (1998) 469–473.
- [4] B. Noheda, D.E. Cox, G. Shirane, R. Guo, B. Jones, L.E. Cross, Stability of the monoclinic phase in the ferroelectric perovskite $\text{PbZr}_{1-x}\text{Ti}_x\text{O}_3$, *Phys. Rev. B* 63 (2000) 014103.
- [5] B. Noheda, J.A. Gonzalo, L.E. Cross, R. Guo, S.E. Park, D.E. Cox, G. Shirane, Tetragonal-to-monoclinic phase transition in a ferroelectric perovskite: the structure of $\text{PbZr}_{0.52}\text{Ti}_{0.48}\text{O}_3$, *Phys. Rev. B* 61 (2000) 8687–8695.
- [6] N. Ichinose, N. Miyamoto, S. Takahashi, Ultrasonic transducers with functionally graded piezoelectric ceramics, *J. Eur. Ceram. Soc.* 24 (2004) 1681–1685.
- [7] I. Franke, K. Roleder, L. Mitoseriu, R. Piticescu, Z. Ujma, High-temperature macroscopic piezoelectricity in Nb-doped $\text{PbZr}_{1-x}\text{Ti}_x\text{O}_3$ ceramics driven by the existence of polar regions, *Phys. Rev. B* 73 (2006) 144114.
- [8] S. Zhang, F. Yu, Piezoelectric materials for high temperature sensors, *J. Am. Ceram. Soc.* 94 (2011) 3153–3170.
- [9] C.-M. Wang, J.-N. Chen, H.-T. Lu, Q. Wang, High Curie temperature bismuth layer-structured ferroelectric calcium bismuth niobate ($\text{CaBi}_2\text{Nb}_2\text{O}_9$) piezoelectric ceramics, *Chin. Sci. Bull.* 66 (2021) 2061–2070.
- [10] B.H. Park, B.S. Kang, S.D. Bu, T.W. Noh, J. Lee, W. Jo, Lanthanum-substituted bismuth titanate for use in non-volatile memories, *Nature* 401 (1999) 682–684.
- [11] H. Yan, H. Zhang, R. Ulic, M.J. Reece, J. Liu, Z. Shen, Z. Zhang, A lead-free high-curie-point ferroelectric ceramic, $\text{CaBi}_2\text{Nb}_2\text{O}_9$, *Adv. Mater.* 17 (2005) 1261–1265.
- [12] G.L. Messing, S. Trolier-McKinstry, E.M. Sabolsky, C. Duran, S. Kwon, B. Brahmaroutu, P. Park, H. Yilmaz, P.W. Rehrig, K.B. Eitel, E. Suvaci, M. Seabaugh, K.S. Oh, Templated grain growth of textured piezoelectric ceramics, *Crit. Rev. Solid State Mater. Sci.* 29 (2004) 45–96.
- [13] Q. Wang, C.-M. Wang, J.-F. Wang, S. Zhang, High performance Aurivillius-type bismuth titanate niobate ($\text{Bi}_3\text{TiNbO}_9$) piezoelectric ceramics for high temperature applications, *Ceram. Int.* 42 (2016) 6993–7000.
- [14] C.-M. Wang, J.-F. Wang, Aurivillius phase potassium bismuth titanate: $\text{K}_{0.5}\text{Bi}_{4.5}\text{Ti}_4\text{O}_{15}$, *J. Am. Ceram. Soc.* 91 (2008) 918–923.
- [15] Q. Wang, E.-M. Liang, C.-M. Wang, High performance bismuth titanate-ferrite piezoelectric ceramics for high-temperature applications, *J. Eur. Ceram. Soc.* 44 (2024) 5080–5087.
- [16] Q. Wang, E.-M. Liang, C.-M. Wang, Origin of strong piezoelectric enhancement in bismuth titanate-ferrite for high-temperature applications, *J. Am. Ceram. Soc.* 107 (2024) 6108–6118.
- [17] Q. Wang, E.-M. Liang, C.-M. Wang, High-performance bismuth titanate-ferrite ($\text{Bi}_3\text{Ti}_3\text{FeO}_{15}$) for high-temperature piezoelectric applications, *J. Am. Ceram. Soc.* 107 (2024) 4811–4823.
- [18] X.-Y. Yu, Q. Wang, H.-L. Li, Y.-J. Wan, E.-M. Liang, C.-M. Wang, Rare-earth praseodymium-substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ exhibiting enhanced piezoelectric properties for high-temperature application, *ChemPhysMater* 3 (2024) 431–439.
- [19] L.-M. He, Q. Wang, X.-N. Zhao, S.-S. Kang, C.-M. Wang, Room temperature ferroelectricity and ferromagnetism in Ni-substituted $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$, *Mater. Res. Express* 10 (2023) 125702.
- [20] X. Zuo, S. Zhu, J. Bai, E. He, Z. Hui, P. Zhang, D. Song, W. Song, J. Yang, X. Zhu, J. Dai, Enhanced multiferroicity and narrow band gap in B-site Co-doped Aurivillius $\text{Bi}_5\text{FeTi}_3\text{O}_{15}$, *Ceram. Int.* 45 (2019) 137–143.
- [21] J.-M. Yan, K. Wang, Z.-X. Xu, J.-S. Ying, T.-W. Chen, G.-L. Yuan, T. Zhang, H.-W. Zheng, Y. Chai, R.-K. Zheng, Large ferroelectric-polarization-modulated

- photovoltaic effects in bismuth layered multiferroic/semiconductor heterostructure devices, *J. Mater. Chem. C* 9 (2021) 3287–3294.
- [22] H. Sun, T. Yao, X. Xie, Y. Lu, Y. Wang, Z. Xu, J. Han, X. Chen, Ni, Eu-Co doping effect on the photocatalytic activity and magnetic recyclability in multifunctional single-phase photocatalysts $\text{Bi}_5\text{FeTi}_3\text{O}_{15}$, *J. Colloid Interf. Sci.* 534 (2019) 499–508.
- [23] C. Moure, L. Lascano, J. Tartaj, P. Duran, Electrical behaviour of $\text{Bi}_5\text{FeTi}_3\text{O}_{15}$ and its solid solutions with $\text{CaBi}_4\text{Ti}_4\text{O}_{15}$, *Ceram. Int.* 29 (2003) 91–97.
- [24] Z. Huang, G.-S. Wang, Y.-C. Li, R.-H. Liang, F. Cao, X.-L. Dong, Electrical properties of (Na,Ce) doped $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics, *Phys. Status Solidi A* 208 (2011) 1047–1051.
- [25] Q. Wang, C.-M. Wang, Enhanced piezoelectric properties of Mn-modified $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ for high-temperature applications, *J. Am. Ceram. Soc.* 103 (2020) 2686–2693.
- [26] J.-N. Chen, Q. Wang, X. Zhao, C.-M. Wang, Significantly enhanced dc electrical resistivity and piezoelectric properties of Tb-modified $\text{CaBi}_2\text{Nb}_2\text{O}_9$ ceramics for high-temperature piezoelectric applications, *J. Am. Ceram. Soc.* 105 (2022) 4815–4826.
- [27] B. Mamatha, A.R. James, P. Sarah, Electromechanical properties of A-Site (LiHo)-modified strontium bismuth titanate ($\text{SrBi}_4\text{Ti}_4\text{O}_{15}$) piezoelectric ceramics, *Ferroelectrics* 413 (2011) 115–122.
- [28] Y. Bai, J. Chen, R. Tian, S. Zhao, Enhanced multiferroic and magnetoelectric properties of Ho, Mn co-doped $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ films, *Mater. Lett.* 164 (2016) 618–622.
- [29] A. Zhang, R. Chu, G. Lu, Y. Zhang, L. Duan, G. Li, Z. Xu, Electrical, luminescent properties and electronic structure of (Ho, Nb) co-doped BNT-based multifunctional ceramics, *Ceram. Int.* 49 (2023) 20799–20807.
- [30] M. Onoe, H. Jumonji, Useful formulas for piezoelectric ceramic resonators and their application to measurement of parameters, *J. Acoust. Soc. Am.* 41 (1967) 974–980.
- [31] X. Du, I.W. Chen, Ferroelectric thin films of bismuth-containing layered perovskites: Part I, $\text{Bi}_4\text{Ti}_3\text{O}_{12}$, *J. Am. Ceram. Soc.* 81 (1998) 3253–3259.
- [32] Q. Wang, Z.-P. Cao, C.-M. Wang, Q.-W. Fu, D.-F. Yin, H.-H. Tian, Thermal stabilities of electromechanical properties in cobalt-modified strontium bismuth titanate ($\text{SrBi}_4\text{Ti}_4\text{O}_{15}$), *J. Alloy. Compd.* 674 (2016) 37–43.
- [33] S. Guan, B. Wang, W. Shi, F. Zhang, Y. Wu, H. Xu, J. Xing, Q. Chen, Realization of high piezoelectric performance in bismuth titanate niobate via B-site mixed-valence doping, *J. Eur. Ceram. Soc.* 43 (2023) 928–938.
- [34] J. Xi, H. Chen, X. Peng, H. Wan, Y. Lin, Z. Tan, J. Xing, Q. Chen, J. Zhu, Z.-G. Ye, Achieving significantly enhanced piezoelectricity in aurivillius ceramics by improving initial polarization and dielectric breakdown strength, *J. Eur. Ceram. Soc.* 43 (2023) 4757–4765.
- [35] H. Yan, Z. Zhang, W. Zhu, L. He, Y. Yu, C. Li, J. Zhou, The effect of (Li,Ce) and (K, Ce) doping in Aurivillius phase material $\text{CaBi}_4\text{Ti}_4\text{O}_{15}$, *Mater. Res. Bull.* 39 (2004) 1237–1246.
- [36] Y. Liu, Y. Zhang, L. Zhu, P. Zheng, W. Bai, L. Li, F. Wen, Z. Xu, L. Zheng, Y. Zhang, Enhanced piezoelectric activity with good thermal stability and improved electrical resistivity in Ta-Mn co-doped $\text{CaBi}_4\text{Ti}_4\text{O}_{15}$ high-temperature piezoceramics, *Ceram. Int.* 46 (2020) 22532–22538.
- [37] X. Jiang, X. Jiang, C. Chen, N. Tu, Y. Chen, B. Zhang, Photoluminescence, structural, and electrical properties of erbium-doped $\text{Na}_{0.5}\text{Bi}_{4.5}\text{Ti}_4\text{O}_{15}$ ferroelectric ceramics, *J. Am. Ceram. Soc.* 99 (2016) 1332–1339.
- [38] C.-M. Wang, J.-F. Wang, S. Zhang, T.R. Shrout, Electromechanical properties of A-site (LiCe)-modified sodium bismuth titanate ($\text{Na}_{0.5}\text{Bi}_{4.5}\text{Ti}_4\text{O}_{15}$) piezoelectric ceramics at elevated temperature, *J. Appl. Phys.* 105 (2009) 094110.
- [39] X. Xie, Z. Zhou, T. Chen, R. Liang, X. Dong, Enhanced electrical properties of NaBi modified $\text{CaBi}_2\text{Nb}_2\text{O}_9$ -based Aurivillius piezoceramics via structural distortion, *Ceram. Int.* 45 (2019) 5425–5430.
- [40] Q. Wei, M. Zhu, M. Zheng, Y. Hou, $\text{Sr}(\text{Zn}_{1/3}\text{Nb}_{2/3})\text{O}_3$ -induced R3c to P4bm transition and large field-induced strain in $0.80(\text{Bi}_{0.5}\text{Na}_{0.5})\text{TiO}_3$ -0.20 SrTiO_3 ceramics, *J. Mater. Res.* 34 (2019) 1210–1218.
- [41] J. Wang, Y. Shen, F. Song, F. Ke, Y. Bai, C. Lu, Effects of oxygen vacancies on polarization stability of barium titanate, *Sci. China Phys. Mech. Astron.* 59 (2016) 634602.
- [42] Q.-M. Zhang, W.Y. Pan, S.J. Jang, L.E. Cross, Domain wall excitations and their contributions to the weak-signal response of doped lead zirconate titanate ceramics, *J. Appl. Phys.* 64 (1988) 6445–6451.
- [43] D. Damjanovic, Contributions to the piezoelectric effect in ferroelectric single crystals and ceramics, *J. Am. Ceram. Soc.* 88 (2005) 2663–2676.
- [44] Q.-M. Zhang, H. Wang, N. Kim, L.E. Cross, Direct evaluation of domain-wall and intrinsic contributions to the dielectric and piezoelectric response and their temperature dependence on lead zirconate-titanate ceramics, *J. Appl. Phys.* 75 (1994) 454–459.
- [45] J.-N. Chen, C. Kang, R.-M. Hou, X. Zhao, C.-M. Wang, Dielectric, ferroelectric, and piezoelectric properties of Gd-modified $\text{CaBi}_2\text{Nb}_2\text{O}_9$ high Curie temperature ceramics, *J. Mater. Res.* 36 (2021) 1086–1096.
- [46] C.-M. Wang, J.-F. Wang, L.-M. Zheng, M.-L. Zhao, C.-L. Wang, Enhancement of the piezoelectric properties of sodium lanthanum bismuth titanate ($\text{Na}_{0.5}\text{La}_{0.5}\text{Bi}_4\text{Ti}_4\text{O}_{15}$) through modification with cobalt, *Mat. Sci. Eng. B* 171 (2010) 79–85.
- [47] C.-M. Wang, L. Zhao, J.-F. Wang, L.-M. Zheng, J. Du, M.-L. Zhao, C.-L. Wang, Cerium-modified Aurivillius-type sodium lanthanum bismuth titanate with enhanced piezoactivities, *Mat. Sci. Eng. B* 163 (2009) 179–183.
- [48] Z. Yue, G. Tan, W. Yang, H. Ren, A. Xia, Enhanced multiferroic properties in Pr-doped $\text{BiFe}_{0.97}\text{Mn}_{0.03}\text{O}_3$ films, *Ceram. Int.* 42 (2016) 18692–18699.
- [49] Z. Huang, S.B. Guo, Z.Y. Cai, Y.C. Li, G.S. Wang, X.L. Dong, Conduction of (Na,Ce) doped $\text{Bi}_5\text{Ti}_3\text{FeO}_{15}$ ceramics after different annealing processes, *Adv. Appl. Ceram.* 112 (2013) 136–139.
- [50] S. Ehara, K. Muramatsu, M. Shimazu, J. Tanaka, M. Tsukioka, Y. Mori, T. Hattori, H. Tamura, Dielectric properties of $\text{Bi}_4\text{Ti}_3\text{O}_{12}$ below the curie temperature, *Jpn. J. Appl. Phys.* 20 (1981) 877–881.
- [51] X. Yang, A.J. Fernández-Carrión, J. Wang, F. Porcher, F. Fayon, M. Allix, X. Kuang, Cooperative mechanisms of oxygen vacancy stabilization and migration in the isolated tetrahedral anion Scheelite structure, *Nat. Commun.* 9 (2018) 4484.
- [52] J.-N. Chen, Q. Wang, H.-T. Lu, X. Zhao, C.-M. Wang, Enhanced electrical properties and conduction mechanism of A-site rare-earth Nd-substituted $\text{CaBi}_2\text{Nb}_2\text{O}_9$, *J. Phys. D.* 55 (2022) 315301.
- [53] X.-P. Jiang, X.-J. Wang, J.-X. Wen, C. Chen, N. Tu, X.-H. Li, Microstructure and electrical properties of Mn-modified bismuth-layer $\text{Na}_{0.25}\text{K}_{0.25}\text{Bi}_{2.5}\text{Nb}_2\text{O}_9$ Ceramics, *J. Alloy. Compd.* 544 (2012) 125–128.
- [54] H. Nagata, S. Matsuzawa, T. Tokutsu, S. Inai, M. Suzuki, Y. Hiruma, T. Takenaka, Temperature dependence of piezoelectric properties on Nd and V co-substituted $\text{Bi}_4\text{Ti}_3\text{O}_{12}$ ceramics for ceramic resonator applications, *Ceram. Int.* 35 (2009) 163–167.